Venus Yan

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EDUCATION

Washington University in St. Louis, St. Louis, MO
MS in Artificial Intelligence Engineering (Intended) — Expected Graduation May 2029
BS in Computer Science — Expected Graduation May 2028

Juniata College, Huntingdon, PA — CS GPA: 3.88/4.0 | General GPA: 3.6/4.0
BS in Mathematics & Computer Science — May 2026

RESEARCH EXPERIENCE

Juniata College, Huntingdon, PA — May 2025-Sep. 2025
Student Researcher for Dataset Distillation (Advised by Dr. Jianyuan Ni)

- Reproduced and evaluated a model-pool-based gradient-matching framework for dataset distillation in PyTorch across 6 neural network architectures, achieving 76% dataset size reduction while increasing Top-1 accuracy by 0.5 percentage points.
- Integrated knowledge distillation using KL divergence and cross-entropy loss to refine cross-architecture generalization across ConvNet, MLP, LeNet, AlexNet, VGG11, ResNet18.

Juniata College, Huntingdon, PA — Mar. 2026-Jun. 2026
Student Researcher for AI Agent Development in Education (Advised by Dr. Shuang Quan)

- Investigated prompt/instruction and interaction strategies for an AI-assisted language learning system, improving children's engagement by 25%+.
- Redesigned agent workflows using finite-state machines and migrated to updated AI APIs to improve response consistency and interaction reliability.

Juniata College, Huntingdon, PA — Jan. 2026-May 2026
Student Researcher for Statistics on Parking Functions and Dyck Paths via Binary Tree Bijections (Advised by Dr. Kristin Camenga)

- Designed a recursive bijection framework connecting parking functions, Dyck paths, and rooted plane binary trees, translating structural properties and statistics across combinatorial representations.
- Developed and proved an associative tree-merging algorithm and derived statistic-preserving correspondences between parking functions and Dyck paths.
- Constructed an explicit transformation on Dyck paths with fixed statistics and used computational enumeration to identify and analyze symmetry patterns in statistic distributions.

TECHNICAL SKILLS

Programming languages: Python, C/C++, Java, SQL, R, JavaScript
ML/Scientific Computing: PyTorch, Scikit Learn, Pandas, NumPy, Matplotlib, Jupyter
Tools: Git/GitHub, Linux, LaTeX, Docker, Arduino
Databases/Cloud: MySQL, AWS, Azure, MongoDB, Oracle
Languages: English(Full professional), Chinese(Native)
Math coursework: Real Analysis, Differential Equations, Mathematical Modeling, Scientific Computing, Abstract Algebra, Linear Algebra, Discrete Structures, Calculus, Statistical Consulting
CS coursework:Intro Machine Learning and Pattern Classification, AI Fundamentals, Algorithms and Analysis, Software Engineering, Database Management Systems, Parallel and Concurrent Programming

RESEARCH PRESENTATION

Liberal Arts Symposium at Juniata College, Huntingdon, PA — Apr. 23, 2026
"How Parking 'Queues' Build Trees: A Visual Story of Combinatorics"
MAA Allegheny Mountain Section Meeting at Penn State, New Kensington, PA — Mar. 27, 2026
"A Constructive Bijection Between Parking Functions, Dyck Paths, and Binary Trees"
Landmark Symposium at Elizabethtown College, Elizabethtown, PA — Jul. 10, 2025

“Model-Pool Driven Dataset Distillation for Architecture-Agnostic Generalization”

Liberal Arts Symposium at Juniata College, Huntingdon, PA Apr. 24, 2025

“Imagining Single-player Tsuro”

SOFTWARE ENGINEERING EXPERIENCE

Juniata College, Huntingdon, PA Jan. 2025-May 2026

Student Developer/Team lead for Juniata Web Development

- Led team of 5 by coordinating tasks for student developers, communicating with campus stakeholders to clarify requirements to align solutions with user needs, ensuring consistent and timely delivery.
- Overhauled backend logic and SQL queries for work order system by merging new user options and streamlining repetitive processes through iterative logic to increase extensibility and maintainability.
- Used JavaScript, HTML/CSS, Microsoft SQL Server to design and maintain web applications.
- Proposed and integrated UI/UX enhancements, including clearer instructions and automated error prevention for incorrect job types, reducing average task completion time by 30%+.